MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10549234

(BOARD, UNIVERSAL)

## 1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the universal board assembly used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-C-50652 | Circuit Card Assemblies, General Specification for |

DRAWINGS

U.S. Army, Frankford Arsenal

| | |
|---|---|
| F10549234 | Board, Universal |

PACKAGING DATA SHEET

| | |
|---|---|
| 10549234 | Packaging of Universal Board |

(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1285

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3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10549234 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. - For the requirements of 3.5.4, 3.5.5 and 3.5.7 logic input, output levels shall be as follows:

Logic 0 — 0.0 ±0.5 Vdc level
Logic 1 — +3.5 ±1.0 Vdc level

3.5.1 Input signals. - Input signals to the universal board shall be T1, $\overline{T1}$, $\overline{T2}$, $\overline{T4}$, $\overline{T5}$ and $\overline{PRF}$ as specified in Figure 1. A positive pulse shall be a change in direct current (dc) level from 0.0 ±0.5 volt (V) to +3.5 ±1.0 V and return to 0 V. A negative pulse shall be a change in dc level from +3.5 ±1.0 to 0.0 ±0.5 V and return to +3.5 V.

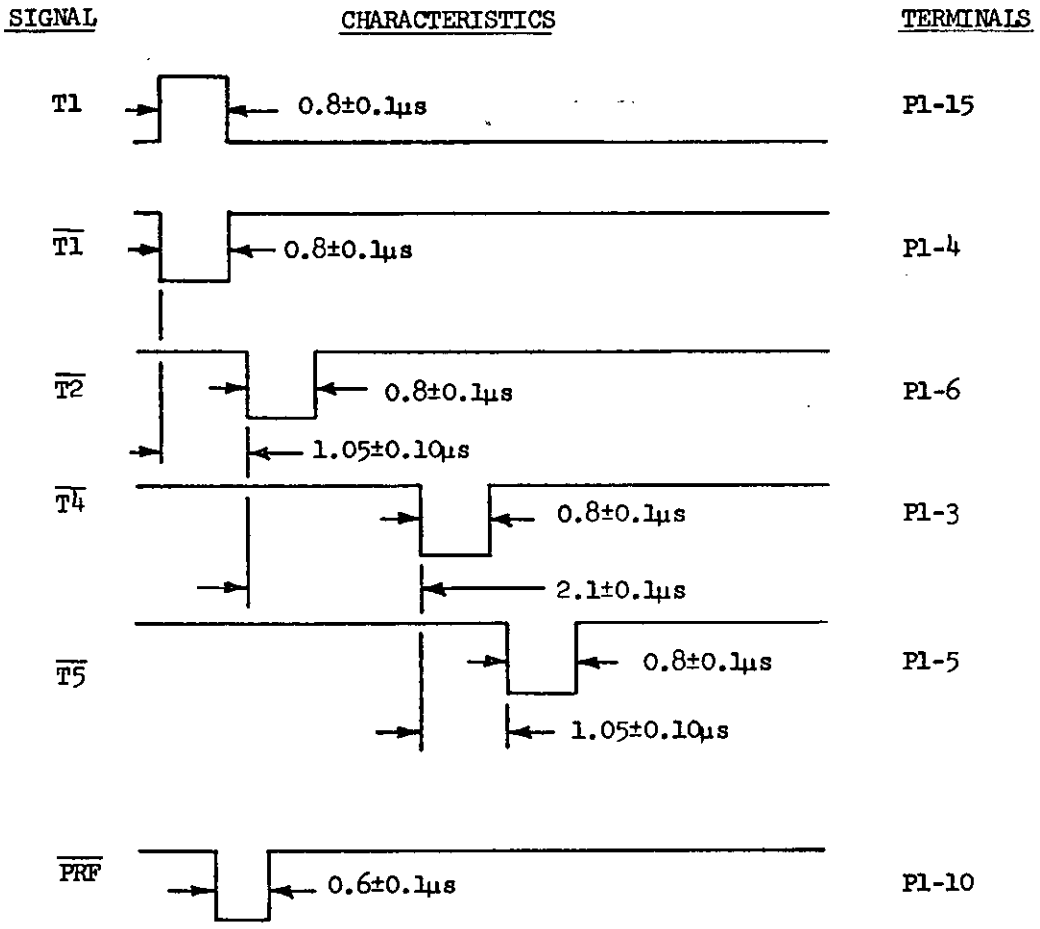
3.5.2 Input power. - The input power applied between terminals P1-30, P1-31, and P1-32 (+5 V) and terminals P1-24, P1-25, and P1-26 (+5 V return) shall be 5.0 ±0.3 volts direct current (Vdc) at 200 milliamperes (mA), regulated; with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.3 E/L TRIG output. - At P1-17 a positive pulse shall be present, having the same characteristics as signal T1 specified in Figure 1 applied to P1-15 except delayed between 1.0 and 1.3 microsecond (μs).

3.5.4 $\overline{R\ RESET}$ output. - With a logic 1 level applied to P1-14, three negative pulses having the same characteristics as signals $\overline{T2}$, $\overline{T4}$ and $\overline{T5}$ specified in Figure 1 shall be present at P1-11. With a logic 0 applied to P1-14, two negative pulses having the same characteristics as signals $\overline{T2}$ and $\overline{T4}$ specified in Figure 1 shall be present at P1-11.



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All pulses at a 20 kHz PRF.

FIGURE 1. INPUT SIGNALS.



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3.5.5 PRF output. - At P1-34, a positive pulse complementary to and having the same characteristics as the $\overline{PRF}$ signal specified in Figure 1 shall be present.

3.5.6 MAN RANGE FREEZE output. - With a logic 1 level applied to P1-7, a logic 0 level shall be present at P1-8. With a logic 0 level applied at P1-7 a logic 1 level shall be present at P1-8.

3.5.7 $\overline{\text{MIN RANGE TRIG}}$ output. - At P1-1, a positive pulse complementary to and having the same characteristics as signal T1 specified in Figure 1 shall be present.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2.1 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table I shall be utilized in the performance of the applicable tests as specified in 4.6.3.



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TABLE I. - TEST EQUIPMENT LIST

| ITEM NUMBER | DESCRIPTION |
|---|---|
| 1. Power supply | 5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity, 200 mA. |
| 2. Data generator (digital word generator) | Four parallel channels, minimum of five selectable bits per channel, with 1/0 content of each bit independently programmed. Data output shall be of return to zero (RZ) with controllable pulse width of 0.8 ±0.1 μs at a 20 kHz PRF. Output pulse amplitude, 3.5 ±1.0 Vdc into a 50-ohm load. |
| 3. Pulse generator | 20 kHz positive pulse; 3.5 ±1.0 Vdc into a 50-ohm load; 0.5 ±0.1 μs wide. Pulse rise and fall times less than 0.1 μs. |
| 4. Oscilloscope | Dual channel; rise time of scope plus measuring probe, 10 ns or less; sweep period, 50 ns/cm or greater; input resistance, 8 megohms or greater; input capacitance, 25 PF or less. |
| 5. Test fixture | As specified in Figure 2. See Table II for test fixture parts list. |



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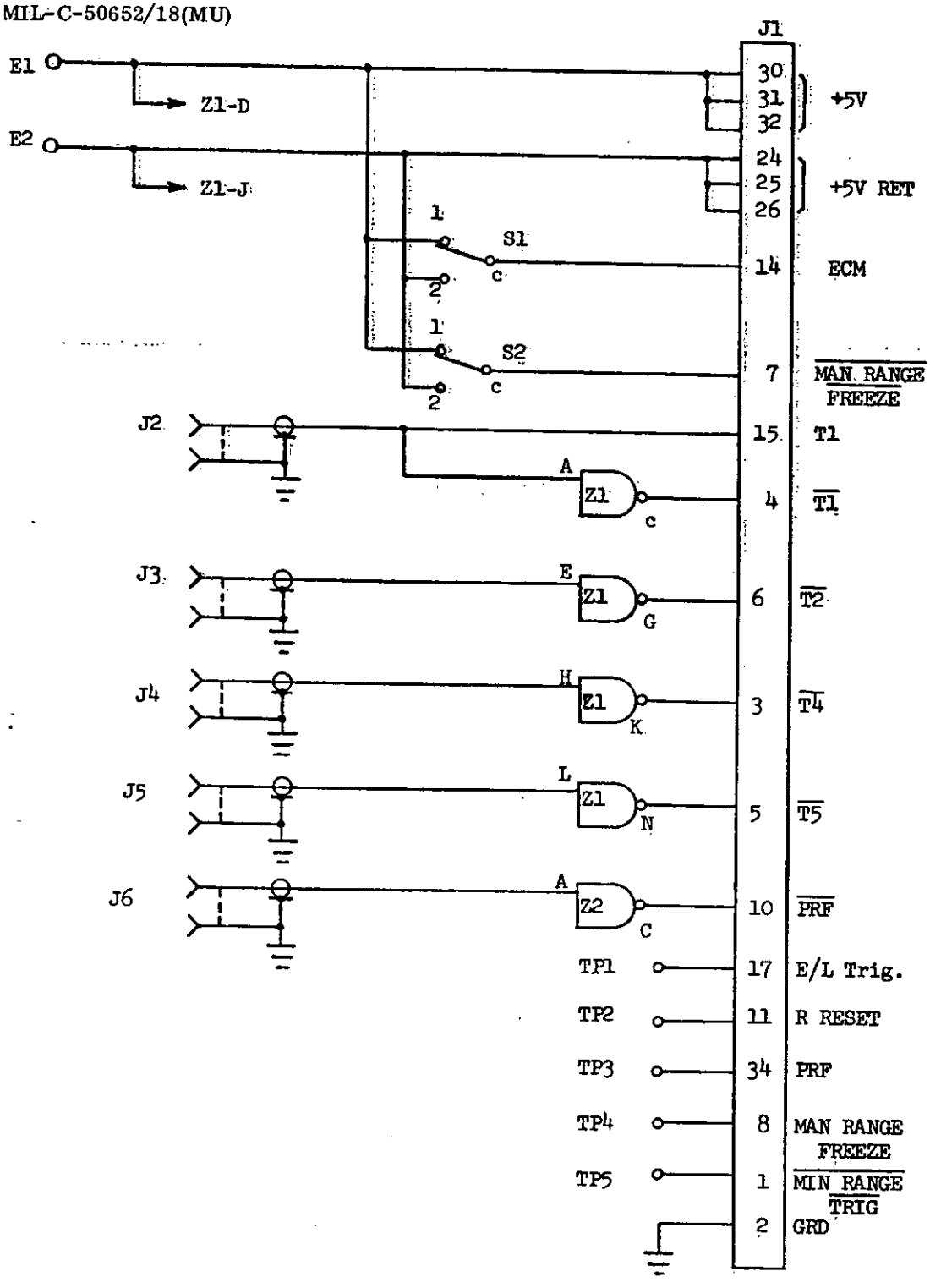



FIGURE 2. TEST FIXTURE SETUP.



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TABLE II. - TEST FIXTURE PARTS LIST

| DESIGNATION | NOMENCLATURE AND DESCRIPTION |
|---|---|
| E1 | Binding post, H.H. Smith 257-R or equivalent. |
| E2 | Binding post, H.H. Smith 257-B or equivalent. |
| J1 | Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp. Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa. |
| J2 thru J6 | Connector, receptacle, UG-25B/U or equivalent. |
| S1 | Switch SPDT. |
| S2 | Same as S1. |
| TP1 thru TP5 | Tip jack, H.H. Smith 1501 or equivalent. |
| Z1 and Z2 | Integrated circuit, SG-140-03 Sylvania or equivalent. |

4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.



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4.6.3 Performance. - The selected universal boards shall be tested as follows: Set up equipment as shown in Figure 3. Use oscilloscope, item 4, for observation of input and output signals specified in 4.6.3.1 through 4.6.3.7. When specified a logic 0 shall be observed as a 0.0 ±0.5 Vdc level and a logic 1 as a +3.5 ±1.0 Vdc level. Switches S1 and S2 on test fixture, item 5, are used to supply logic level input signals specified in 4.6.3.4 and 4.6.3.6. Switches supply a logic 1 in position 1 and a logic 0 in position 2.

4.6.3.1 Input signals. - Using data generator, item 2, set up and apply signal T1 in accordance with Figure 1 and signals T2, T4, and T5. Signals T2, T4, and T5 are complementary to signals $\overline{T2}$, $\overline{T4}$, and $\overline{T5}$ shown in Figure 1 and shall have the same characteristics.

4.6.3.2 Input power. - Using item 1, apply input power as specified in 3.5.2.

4.6.3.3 E/L TRIG output. - At TP1 on the test fixture, item 5, observe the E/L TRIG output signal as specified in 3.5.3.

4.6.3.4 $\overline{\text{R RESET}}$ output. - Using S1 on the test fixture, item 5, to apply logic levels as specified in 3.5.4. Observe at TP2 the $\overline{\text{R RESET}}$ output signal as specified in 3.5.4.

4.6.3.5 PRF output. - At TP3 on the test fixture, item 5, observe the PRF output signal as specified in 3.5.5.

4.6.3.6 MAN RANGE FREEZE output. - Using S2 on the test fixture, item 5, apply logic levels as specified in 3.5.6. Observe at TP4 the MAN RANGE FREEZE output signal as specified in 3.5.6.

4.6.3.7 $\overline{\text{MIN RANGE TRIG.}}$ output. - At TP5 on the test fixture, item 5, observe the $\overline{\text{MIN RANGE TRIG}}$ output signal as specified in 3.5.7.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10549234. The level of protection shall be as specified in the procurement document.



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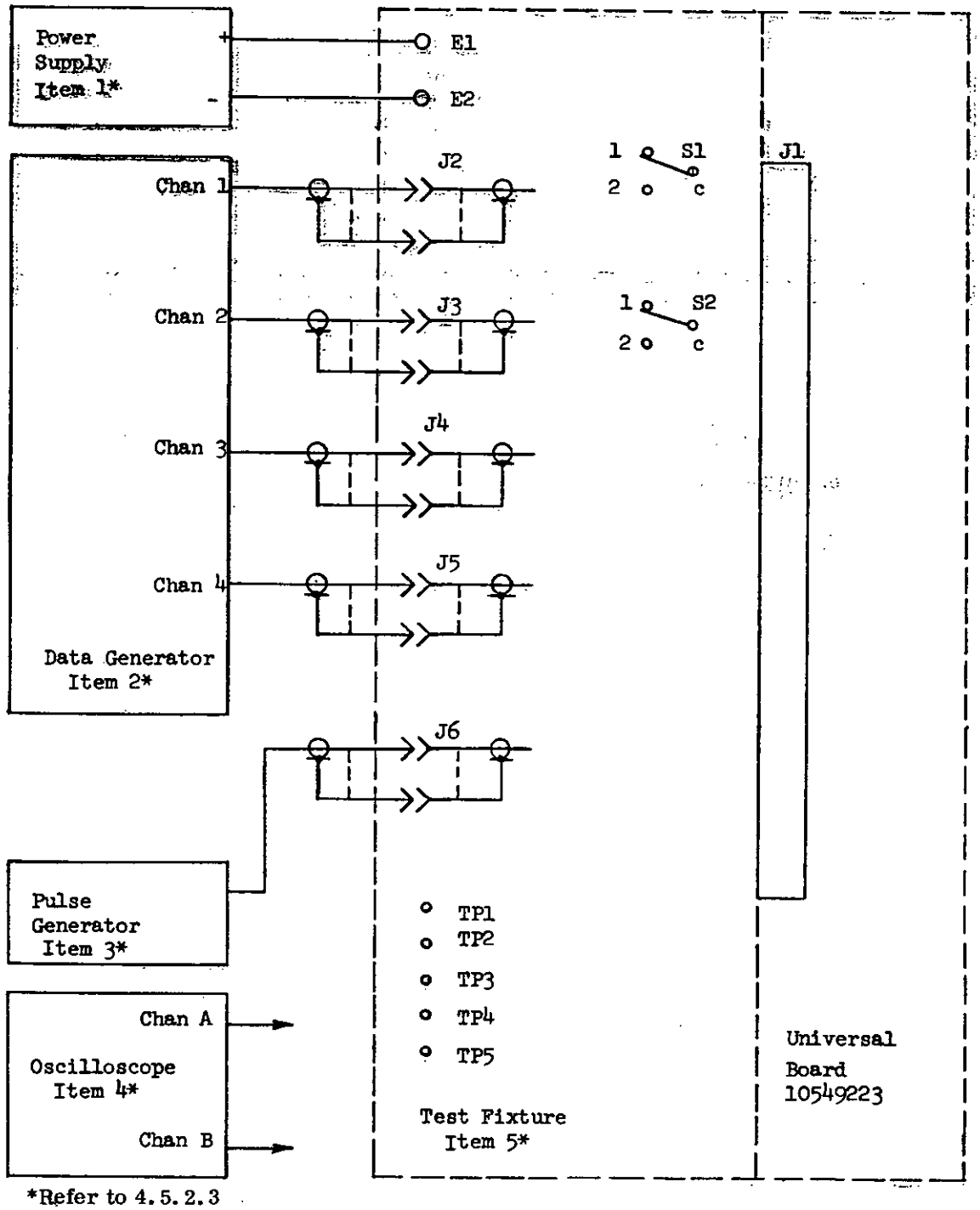



FIGURE 3. UNIVERSAL BOARD TEST SETUP.



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6. NOTES

6.1 Intended use. - The universal board assembly is used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The universal board provides five signal outputs: the E/L TRIG output, the R RESET, the PRF, the MAN RANGE FREEZE, and the MIN RANGE TRIG.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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